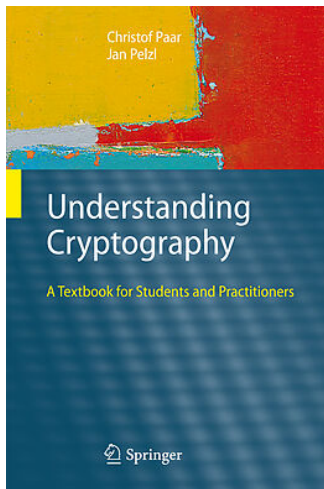




HA NOI UNIVERSITY OF SCIENCE AND TECHNOLOGY
SCHOOL OF INFORMATION AND COMMUNICATION TECHNOLOGY

Introduction to Cryptography and Security

Introduction



- 1 Introduction
- 2 Symmetric Cryptography
- 3 Cryptanalysis
- 4 Modular Arithmetic and More Historical Ciphers
- 5 Lessons Learned

World War II

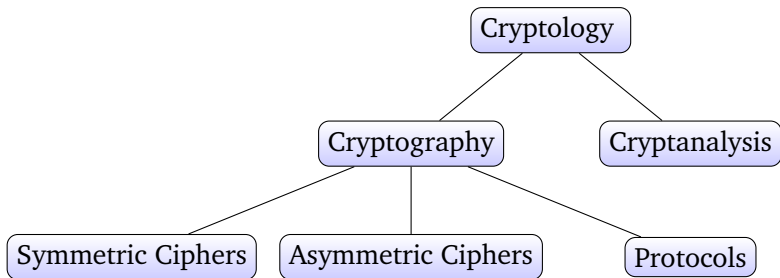
German Enigma encryption machine



Back to about 2000 B.C

Scytale of Sparta



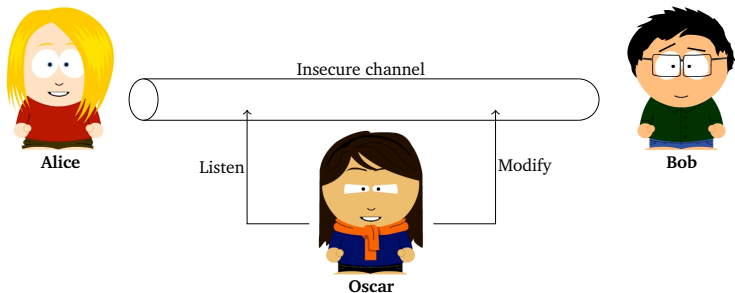


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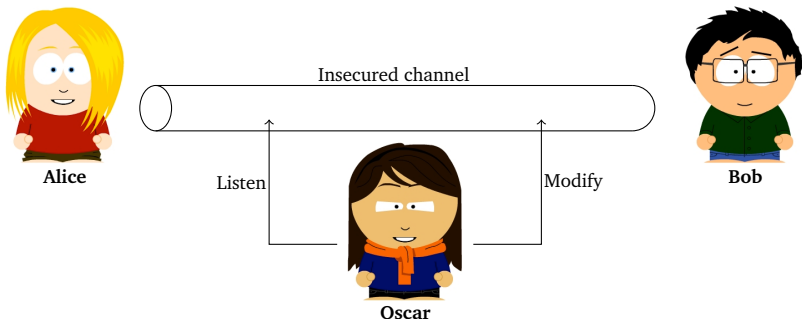
Goal: Confidentiality of transmitted (or stored) message.

Characters in the game :

- Alice, Bob are “good guys” (by Wikipedia)
- Oscar is “eavesdropper”, “adversary”



Cryptography Approach



- Bob knows a **key** k that Oscar doesn't (Oscar know the system).
- Alice can **encrypt** x such that knowledge of k allows for decryption.
- Oscar sees ciphertext y , but learns **nothing** about x .

- x, m are plaintext or message;
- y, c are ciphertexts;
- k is the key;
- Enc is the encryption function;
- Dec is the decryption function;
- Gen is the key generation function.

Symmetric Cryptography

Alice & Bob both know key k . **shared symmetric key**

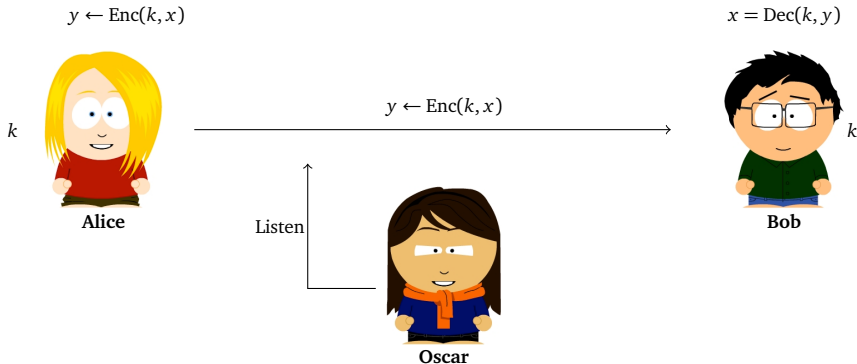
Algorithms:

$k \leftarrow \text{Gen}(1^\lambda)$ generate key of length λ
 $y \leftarrow \text{Enc}(k, x)$ encrypt message x with key k , result is y
 $x = \text{Dec}(k, y)$ decrypt y using k to obtain x .

Setup:

- Someone (may be Alice or Bob) computes $k \leftarrow \text{Gen}(1^\lambda)$.
- and ensures that Alice & Bob both have k (and Oscar doesn't)
(How?!)

Symmetric-key cryptosystem



- Can Oscar know the encryption and decryption functions?
- Yes. She knows.
- Why not hide the encryption and decryption functions?
- Because it's safer to make the functions public!

A cryptosystem should be secure even if the attacker (Oscar) knows all details about the system, with the exception of the secret key. In particular, the system should be secure when the attacker knows the encryption and decryption algorithms.

Remark: Kerckhoffs' Principle is counterintuitive.

NaCl (Networking and Cryptography library)

- NaCl (pronounced "salt") is a new easy-to-use high-speed software library for network communication, encryption, decryption, signatures, etc.
- NaCl's goal is to provide all of the core operations needed to build higher-level cryptographic tools.
- NaCL in Wikipedia
[https://en.wikipedia.org/wiki/NaCl_\(software\)](https://en.wikipedia.org/wiki/NaCl_(software))
- NaCl was created by the mathematician and programmer Daniel J. Bernstein.

Example

$A \rightarrow k$

$B \rightarrow d$

$C \rightarrow w$

...

For instance, the pop group ABBA would be encrypted as kddk.

What is the key k of this cipher?

Consider the ciphertext that is encrypted by the substitution cipher.

```
1 iq ifcc vqqr fb rdq vfllecq na rdq cfjwhwz hr bnnb  
2 hcc hwwhbsqvqbrc hwq vhlq
```

Question

- Can you guess what the plaintext is?
- Is the substitution cipher secure?

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First Attack: Brute-Force or Exhaustive Key Search

- Let (x, y) denote the pair of plaintext and ciphertext,
- and let $K = \{k_1, \dots, k_n\}$ be the key space of all possible keys k_i .
- A brute-force attack now checks for every $k_i \in K$ if

$$\text{Dec}(k_i, y) = x.$$

- If the equality holds, a possible correct key is found; if not, proceed with the next key.

Brute-Force Attack for Substitution Cipher

Question

What is the key space of the cipher?

Second Attack: Letter Frequency Analysis

The major weakness of the substitution cipher is that each plaintext symbol always maps to the same ciphertext symbol.

Example

$A \rightarrow k$

$B \rightarrow d$

$C \rightarrow w$

...

For instance, the pop group ABBA would be encrypted as kddk.

Second Attack: Letter Frequency Analysis

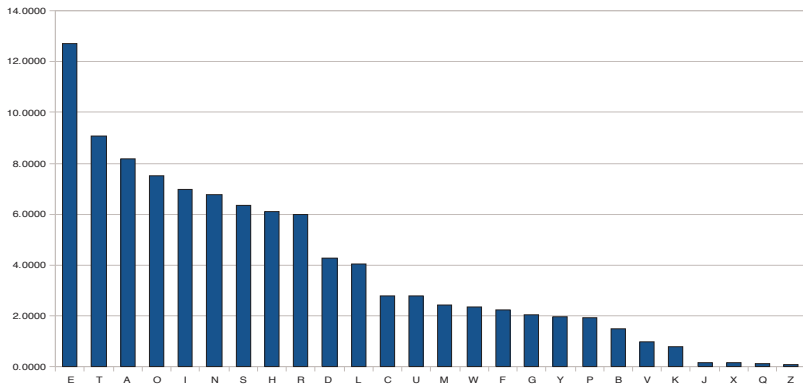
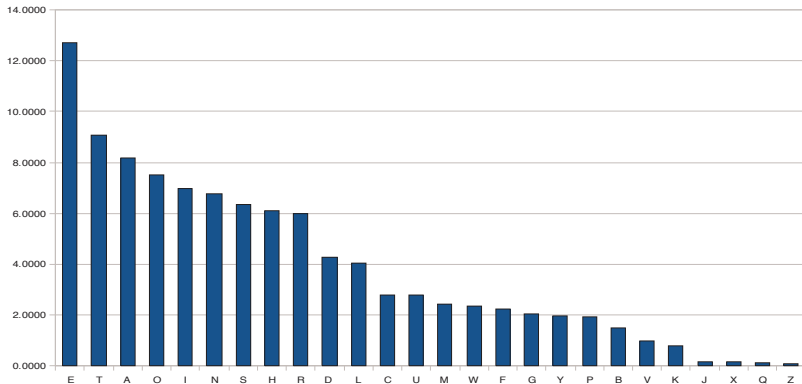


Figure: Relative letter frequencies of the English language

Exercise: Decrypt the ciphertext

```
1 iq ifcc vqqr fb rdq vfllecq na rdq cfjwhwz hr bnnb
2 hcc hwwhbsqvqbre hwq vhlq
```



- Oscar chooses x_1, x_2 (same length).
- She receives either

$$y_1 = \text{Enc}(k, x_1) \quad \text{or} \quad y_2 = \text{Enc}(k, x_2).$$

- Oscar should not be able to figure out which message was encrypted.

$\Rightarrow$ No information about x is leaked from $y = \text{Enc}(k, x)$.

- Key: $k \in \{0, 1\}^n$ chosen uniformly at random.
- Encryption: $y = x \oplus k$
- Decryption: $x = y \oplus k$

x	k	$x \oplus k$
0	0	0
0	1	1
1	0	1
1	1	0

Example

$$x = 1011, \quad k = 1100$$

$$y = x \oplus k = 0111$$

Semantic Security of OTP: Example

- Oscar chooses:

$$x_1 = 0000, \quad x_2 = 1111$$

- Now suppose Oscar receives ciphertext $y = 1010$.
- Can Oscar figure out whether x_0 or x_1 was encrypted to produce ciphertext y ?

Question

Assume you know the ciphertext of the message
“ $x = 1011010011$ ” using one-time pad encryption is

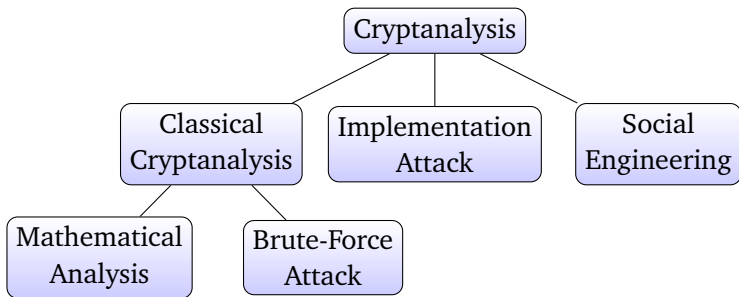
$$y = 1100110010$$

(the plaintext is a 10-bit binary string and the ciphertext is written as a 10-bit binary string). What is the ciphertext of the message
“ $x' = 1011011110$ ” with the same OTP key?

- **Known ciphertext:** the adversary has access only to the ciphertext.
- **Known ciphertext/plaintext pairs:** the adversary has access to both the ciphertext and the corresponding plaintext.
- **Chosen plaintext:** the adversary can choose arbitrary plaintexts to be encrypted and then obtain the corresponding ciphertexts.
- **Chosen ciphertext:** Chosen plaintext and the adversary can choose arbitrary ciphertexts to be decrypted and then obtain the corresponding plaintexts.

Question: Which of these attacks is the most powerful?

Overview of Cryptanalysis



Types of Cryptanalysis: A Diverse Toolkit

- **Classical Cryptanalysis:** Exploits the mathematical structure of the encryption algorithm itself.
- **Brute-Force Attacks:** Exhaustively trying all possible keys until the correct one is found.
- **Implementation Attacks:** Targeting weaknesses in the implementation of a cryptosystem, such as timing differences, power **consumption** patterns, or code vulnerabilities.
- **Social Engineering Attacks:** Manipulating people to reveal secrets or gain access to sensitive information.

Key Length: A Measure of Strength

Key length	Security estimation
56–64 bits	short term: a few hours or days
112–128 bits	long term: several decades in the absence of quantum computers
256 bits	Long term: several decades, even with quantum computers that run the currently known quantum computing algorithms

Key Lengths: A Matter of Context

Symmetric crypto algorithms

- The discussion of key lengths for symmetric crypto algorithms is only relevant if a brute-force attack is the best known attack.

Symmetric vs. Asymmetric

- The key lengths for symmetric and asymmetric algorithms are dramatically different.
- For instance, an 80-bit symmetric key provides roughly the same security as a 1024-bit RSA (RSA is a popular asymmetric algorithm) key.

- It is hard to know what kinds of computers will be available in the year 2030.
- Moore's Law predicts a doubling of computing power every 18 months, while costs remain constant.
- This exponential growth impacts cryptography: attacks become faster and cheaper over time.
- Cryptography must adapt and evolve to stay ahead of this exponential trend.

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- We consider the set of the nine numbers:

$$\{0, 1, 2, 3, 4, 5, 6, 7, 8\}$$

- We can do regular arithmetic as long as the results are smaller than 9. For instance:

$$2 \times 3 = 6$$

$$4 + 4 = 8$$

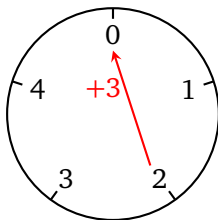
- But what about $8 + 4$? We can use modular arithmetic to simplify the result.

$$8 + 4 \equiv ? \pmod{9}$$

Modular Arithmetic in $\mathbb{Z}_5$

+	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	0
2	2	3	4	0	1
3	3	4	0	1	2
4	4	0	1	2	3

$\times$	0	1	2	3	4
0	0	0	0	0	0
1	0	1	2	3	4
2	0	2	4	1	3
3	0	3	1	4	2
4	0	4	3	2	1



Example

How to calculate $3^8 \bmod 7$?

1 way: $3^8 = 6561 \equiv 2 \bmod 7$

2 way: $3^8 = 3^4 \cdot 3^4 = 81 \cdot 81 \bmod 7$
 $= 4 \cdot 4 \bmod 7$
 $= 16 \bmod 7$
 $= 2 \bmod 7$

Definition (Multiplicative Inverse)

For $a \in \mathbb{Z}_n$, an element $a^{-1} \in \mathbb{Z}_n$ is called the **multiplicative inverse** of a if

$$a \times b \equiv 1 \pmod{n}.$$

Inverses in $\mathbb{Z}_5$

a	$a^{-1} \pmod{5}$
1	1
2	3
3	2
4	4
0	no inverse

Division in $\mathbb{Z}_5$

$/$	1	2	3	4
0	0	0	0	0
1	1	3	2	4
2	2	1	4	3
3	3	4	1	2
4	4	2	3	1

Definition (Multiplicative Inverse)

Let $a, m \in \mathbb{Z}$ and $m > 0$. The multiplicative inverse of a modulo m is an integer a^{-1} such that

$$a \cdot a^{-1} \equiv 1 \pmod{m}.$$

- An element $a \in \mathbb{Z}$ has a multiplicative inverse a^{-1} if and only if $\gcd(a, m) = 1$.
- If $\gcd(a, m) = 1$, then a and m are said to be relatively prime or coprime.

- Does the multiplicative inverse of 15 exist in $\mathbb{Z}_{26}$?
- Because $\gcd(15, 26) = 1$, the inverse must exist. In fact, the inverse is 7 since

$$7 \cdot 15 \equiv 1 \pmod{26}.$$

- On the other hand, the multiplicative inverse of 14 does not exist in $\mathbb{Z}_{26}$.

Not Every Element Has an Inverse

Exercise

Find the multiplicative inverse of every element in $\mathbb{Z}_{12}$.

a	0	1	2	3	4	5	6	7	8	9	10	11
a^{-1}												

Hint: a has an inverse in $\mathbb{Z}_{12}$ iff $\gcd(a, 12) = 1$.

Encoding of Letters for the Shift Cipher

A	B	C	D	E	F	G	H	I	J	K	L	M
0	1	2	3	4	5	6	7	8	9	10	11	12
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
13	14	15	16	17	18	19	20	21	22	23	24	25

Table: Encoding of letters for the shift cipher

Definition (Shift Cipher)

Let $x, y, k \in \mathbb{Z}_{26}$.

- Encryption: $e_k(x) \equiv (x + k) \pmod{26}$.
- Decryption: $d_k(y) \equiv (y - k) \pmod{26}$.

Example

- Let the key be $k = 17$, and the plaintext is:

$$\text{ATTACK} = x_1, x_2, \dots, x_6 = 0, 19, 19, 0, 2, 10.$$

- The ciphertext is then computed as:

$$y_1, y_2, \dots, y_6 = 17, 10, 10, 17, 19, 1 = \text{rkkrtb}.$$

Definition (Affine Cipher)

Let $x, y, a, b \in \mathbb{Z}_{26}$.

- Encryption: $e_k(x) \equiv (a \cdot x + b) \pmod{26}$.
- Decryption: $d_k(y) \equiv (a^{-1} \cdot (y - b)) \pmod{26}$.

with the key: $k = (a, b)$, which has the restriction: $\gcd(a, 26) = 1$.

The decryption is easily derived from the encryption function:

$$a \cdot x + b \equiv y \pmod{26}$$

$$a \cdot x \equiv (y - b) \pmod{26}$$

$$x \equiv a^{-1} \cdot (y - b) \pmod{26}$$

Example

Let the key be $k = (a, b) = (9, 13)$, and the plaintext be ATTACK:

$$x_1, x_2, \dots, x_6 = 0, 19, 19, 0, 2, 10$$

The ciphertext is computed as:

$$y_1, y_2, \dots, y_6 = 13, 2, 2, 13, 5, 25 = \text{nccnfz}$$

For decryption, the inverse of a needs to be determined, which turns out to be $a^{-1} = 3$.

Exercise

This problem deals with the affine cipher where the key is given as $a = 7$ and $b = 22$.

- 1 Decrypt the text below:

falszztysyjzyjkywjrztyjztyynaryjkyswarztyegyyj

- 2 Who wrote the line?

- Is the affine cipher secure? No!
- The key space is only a bit larger than in the case of the shift cipher:

$$\begin{aligned}\text{Key space} &= (\text{\#values for } a) \cdot (\text{\#values for } b) \\ &= 12 \cdot 26 = 312\end{aligned}$$

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- Never ever develop your own cryptographic algorithm unless you have a team of experienced cryptanalysts checking your design.
- Do not use unproven cryptographic algorithms (i.e., symmetric ciphers, asymmetric ciphers, hash functions) or unproven protocols.
- Attackers always look for the weakest point of a cryptosystem. For instance, a large key space by itself is no guarantee of a cipher being secure; the cipher might still be vulnerable against analytical attacks.

- Key lengths for symmetric algorithms in order to thwart exhaustive key-search attacks are:
 - 64 bits: insecure except for data with extremely short-term value.
 - 112–128 bits: long-term security of several decades, including attacks by intelligence agencies unless they possess quantum computers. Based on our current knowledge, attacks are only feasible with quantum computers (which do not exist but might become reality in 1–2 decades).
 - 256 bits: as above, but possibly secure against attacks by quantum computers.
- Modular arithmetic is a tool for expressing historical encryption schemes, such as the affine cipher, in a mathematically elegant way and provides the fundamental basis for many modern cryptographic schemes.



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Thank you!



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